

# From GRAPHENE To BOROPHENE Electronics



**Dr S.S. Verma**  
is a professor  
at Department  
of Physics, Sant  
Longowal Institute  
of Engineering and  
Technology, Sangrur,  
Punjab

**T**he discovery of graphene in 2004 by two researchers, Prof. Andre Geim and Prof. Kostya Novoselov at the University of Manchester (UK), created quite an excitement. This pioneering work won the duo *Nobel Prize in Physics* in 2010.

However, there are some inherited limitations of graphene as a wonder material, forcing scientists to look for an alternative. In this quest, they have discovered a new two-dimensional (2D) material called 'borophene' that is thought to exhibit electronic properties better than graphene. This article dwells on the journey from graphene to borophene.

## Graphene

Graphene is a one-atom thin sheet of carbon atoms arranged in a hexagonal format or a flat monolayer of carbon atoms that are tightly packed into a 2D honeycomb lattice. Owing to exceptional strength, flexibility and conductance, it is expected to shape almost all aspects of future technologies.

Researchers around the world have found that graphene has promise for applications in electronics ranging from displays to random-access memory. Noted for the speed at which electrons pass

through it, graphene has already proved to be a miracle electronic material with properties suitable for design of strong, fast, flexible and transparent electronic devices, printed graphene in paper electronics, and car batteries.

## Its limitations

With applications touching humans in every quartet of life, graphene seems to have limitless capabilities, though with the traditional challenge of its production cost. In 2013, a

graphene flake the size of one micrometre cost more than \$1000. Obviously, this is why graphene is one of the most expensive materials on Earth.

However, upscaling of graphene production is one of the most researched subjects and we are not far away from attaining scalability of the graphene production process. Researchers have devised a method for producing graphene-treated silver nanowires, which could significantly reduce production costs for nanowire-based displays. This could mean that graphene offers a real alternative to indium-tin oxide in flexible low-cost touchscreen displays.

Nevertheless, some of the major limitations of graphene include:

1. High bending stiffness for devices that also need to stretch, compress or even twist.
2. Band gap not suitable to simply get the characteristics of switch on and off.
3. High susceptibility to oxidative environments when used as a catalyst
4. Some toxic qualities. Scientists have discovered that graphene features jagged edges that can easily pierce cell membranes, allowing it to enter into cells and disrupt normal functions.

## Borophene and its properties

Scientists are interested in two-dimensional materials mainly for their unique electronic properties. Borophene is an unusual material because it shows many metallic properties at nanoscale. Even though three-dimensional, or bulk, boron is nonmetallic and semiconducting.

Recently, researchers successfully synthesised borophene. The outcome was not only similar to graphene but also much lighter and stronger, contributing to the novel desirable properties.

Researchers showed that borophene has a record high flexibility, a higher stiffness-to-weight ratio and higher ideal strength than

*Flexible and transparent electronics (Image courtesy: Berkeley Lab, USA)*

